

CALCULUS AND DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: 75 | Total Questions: 3 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	Total
Marks	25	25	25	75

Question 1**[25 Marks]**

Let $f(x,y) = x^2 \sin(y) + y^2 \cos(x)$. \ (a) Compute the gradient $\nabla f(x,y)$. \ (b) Evaluate the gradient at the point $P((\pi/4), (\pi/3))$. \ (c) Find the directional derivative of f at P in the direction of the vector $\mathbf{v} = (2,1)$. \ (d) Hence, determine the unit vector in the direction of maximum increase of f at P and the maximum rate of change.

Question 2**[25 Marks]**

Solve the first-order linear differential equation \ $(dy/dx) + y \tan x = \sin(2x)$, $y(0)=1$. \ (a) Find the integrating factor. \ (b) Obtain the general solution. \ (c) Apply the initial condition to determine the particular solution. \ (d) Sketch the solution curve on the interval $[0, \pi/2)$.

Question 3**[25 Marks]**

Consider the second order homogeneous differential equation $y'' + 4y' + 13y = 0$, $y(0)=2$, $y'(0)=0$. \ (a) Write the characteristic equation and find its roots. \ (b) Write the general solution in terms of real functions. \ (c) Apply the initial conditions to obtain the unique solution. \ (d) Describe the behaviour of the solution as $t \rightarrow \infty$ and sketch the graph for $t \in [0,5]$.
